SECTION 933

ADMIXTURES FOR PORTLAND CEMENT CONCRETE

933.01 Scope - This Section covers general requirements for air entraining and chemical admixtures and Specification requirements for pozzolan admixtures.

933.02 Applicable standards. The standards listed in Table 933-A shall apply unless specified otherwise herein or in the Contract. All referenced standards shall be to the current editions at time of Award.

TABLE 933-A Applicable Standards

ASTM-C 260	Air-entraining Admixtures for Concrete
ASTM-C 494	Chemical Admixtures for Concrete
CAN/CSA A3000	Cementitious Materials for Use in Concrete
CAN/CSA A23.1	Concrete Materials and Methods of Concrete Construction

933.03 Air-Entraining and Chemical Admixtures

a) Air-entraining admixtures shall conform to the requirements of ASTM Standard C 260, Air-Entraining Admixtures for Concrete.

b) Chemical Admixtures shall conform to the requirements of ASTM Standard C 494, Chemical Admixtures for Concrete.

933.04 Pozzolan Admixtures

933.04.01 Definitions) Type N Pozzolan - A natural pozzolan consisting of siliceous or alumino-siliceous material in finely divided form and in the presence of moisture chemically reacts at ordinary room temperatures with calcium hydroxide, released by the hydration of portland cement, to form compounds possessing cementing properties.

ii) Type F Pozzolan - A finely divided residue that results from the combustion of pulverized coal and that is carried from the combustion chamber of a furnace by exhaust gasses.

933.04.02 General - When types N or F pozzolans are used in concrete either as a cement replacement, addition, or both, they shall contribute to the beneficial physical and chemical properties of the hardened concrete through pozzolanic activity. Before a pozzolan is accepted for use in concrete, pozzolan shall first meet the physical and chemical requirements herein specified.

If suitable, the pozzolan shall then be evaluated upon its performance in laboratory concrete trial mixes at various usage rates. Any pozzolan, which does not produce the specified effects or produces adverse effects to concrete shall not be used.

933. 04.03 Sampling and Testing - Types N and F pozzolans shall be sampled and tested for chemical and physical properties in accordance with ASTM-C 311 "Sampling and Testing of Fly Ash or Natural Pozzolans for use as a Mineral Admixture in Portland Cement Concrete", unless otherwise specified in the Contract.

933. 04.04 Chemical Requirements - The chemical requirements for types N and F pozzolans shall be as shown in Table 933-B.

933. 04.05 Physical Requirements

i) Fineness - The fineness of either types N or F pozzolan will be determined in accordance with ASTM C 311 Section 21, "Fineness, Amount Retained When Wet-Sieved on a 45 μm (No. 325) Sieve" and/or ASTM C 204 for "Fineness of Hydraulic Cement by Air Permeability Apparatus". In this latter procedure, fineness shall be determined by use of the Blaine permeability apparatus except that pozzolan will be used in place of cement.

The degree of fineness of either type N or F pozzolan shall be that which produces the optimum pozzolanic activity with the Portland cement with which the pozzolan is to be used. The pozzolanic activity index shall however not be less than 75% of the control at 28 days. After six months storage, pozzolans shall be retested for

fineness. The fineness requirements shall be as specified in the Design and the Contract.

ii) Water Requirements - Water requirements of mortar made with the pozzolan in accordance with ASTM C 311 Section 33 shall not be greater than 115% of water used in the control mix.

iii) Drying Shrinkage - Increase in drying shrinkage of mortar bars made and tested in accordance with ASTM C 311 Section 22 shall not be greater than 0.03%.

iv) Soundness - When tested for soundness in accordance with ASTM C 311 the expansion or contraction after autoclaving shall not be greater than 0.08%.

933.04.06 Performance Evaluation of Types N or F Pozzolans in Concrete

i) Only pozzolans which meet the physical and chemical requirements as specified in the Contract shall be evaluated for use in concrete.

ii) Evaluation of pozzolans in concrete shall be made with laboratory concrete trial mixes preferably with concrete ingredients to be used in the Work where the pozzolanic concrete is to be used.

iii) Evaluation shall be made by comparing test results of the pozzolanic mixes with test results of similar concrete mixes without pozzolan.

iv) Tests, which the Quality Manager or the Ministry Representative may specify to evaluate the performance of a pozzolan in a specific concrete, are given in Table 933-C.

933. 04.07 Basis of Evaluation - Types N and F pozzolans shall be evaluated in concretes by comparing results of tests performed on "job design" portland cement concrete with similar class portland cement concrete without pozzolan. Types N and F pozzolans will only be considered for acceptance when they produce equal or better test results than those obtained with similar class concrete without pozzolan. When accepted, types N and F pozzolans shall only be used at dosage rates determined from test results and accepted by the Designer and Ministry Representative.

TABLE 933-B CHEMICAL REQUIREMENTS FOR TYPES N AND F POZZOLANS

CHEMICAL REQUIREMENTS:	POZZOLAN TYPE	
	N	F
Silicon dioxide (SiO_2) plus aluminum oxide (Al_2O_3) plus iron oxide (Fe_2O_3), minimum %	70.0	70.0
Sulphur trioxide (SO_3), maximum %	4.0	5.0
Moisture content, maximum %	3.0	3.0
Loss on ignition, maximum %	10.0	12.0
Magnesium oxide (MgO), maximum %	5.0	5.0
Available alkalies, as Na_2O, maximum %	1.5	1.5

TABLE 933-C TESTS FOR EVALUATING THE PERFORMANCE OF A POZZOLAN IN A SPECIFIC CONCRETE

QUALITY REQUIREMENTS	TITLE OF TEST	TEST DESIGNATION
Air Entrainment	Limits of Amount of Air-Entraining Admixture in Concrete	ASTM C 311 Section 23
Bleeding	Bleeding of Concrete	ASTM C 232
Setting Time	Time of Setting of Concrete Mixtures by Penetration Resistance	ASTM C 403
Compressive Strength	Concrete Test Specimens Making and Curing in the Laboratory	ASTM C 192
Flexural Strength	Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading)	ASTM C 78
Drying Shrinkage	Length Change of Hardened Cement Mortar and Concrete	ASTM C 157
Thermal Volume Change	Early Volume Change of Cementitious Mixtures	ASTM C 827
Alkali Reaction	Effectiveness of Mineral Admixtures in Preventing Excessive Expansion of Concrete Due to the Alkali-Aggregate Reaction	ASTM C 441
Resistance to Freezing and Thawing	Resistance of Concrete to Rapid Freezing and Thawing	ASTM C 666
Resistance to De-icing Chemicals	Scaling Resistance of Concrete Surfaces Exposed to De-icing Chemicals	ASTM C 672

SECTION 941

PRECAST REINFORCED CONCRETE BARRIERS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

941.01 Scope - This Section covers the quality and manufacture of precast reinforced concrete roadside and median barriers for highway and other off-highway traffic confinement use.

The concrete traffic barrier units shall be supplied in the sizes and types as required by the Purchase Order, Work Order or Design in strict conformity with this Section and pertinent Standard SS Drawings of the SP941 Series.

941.02 Concrete Quality

a) Concrete quality shall conform to CSA Standard CAN3-A23.1-M except where amended hereafter.

b) A compressive strength test result is defined as the average of the strengths of three 28 day compressive test cylinder breaks with standard cylinders of 150 mm diameter and 300 mm high size.

c) The strength level of the concrete represented by the test shall be considered satisfactory if the test result equals or exceeds 30 MPa and no individual cylinder strength is less than 27 MPa. If this condition is not met, the concrete will be considered to have failed the strength requirements. No other form of testing to prove the relative strength at a later date will be allowed without the approval of the Ministry Representative.

d) Cylinders shall be cast by the Contractor at the time of placing concrete. Frequency of testing will one set of four cylinders per day or per 50 m^3, whichever is more frequent.

e) Calcium chloride or admixtures containing calcium chloride shall not be used in the concrete.

f) Concrete shall meet the following additional requirements:

i) Minimum cement content of 320 kg per cubic metre.

ii) Maximum water/cement ratio of 0.45.

iii) Coarse aggregate of a nominal maximum size not exceeding 28 mm.

iv) Slump of 50 mm ± 20 mm.

v) Entrained air of 5 to 8%.

941.03 Reinforcing Steel, Attachment Hardware & Miscellaneous Items

a) Welded steel wire mesh reinforcement shall be supplied and installed in each section as shown on the Standard SS Drawings, and in accordance with DBSS 412. Additional reinforcement may be installed to assist handling during the precasting operations but shall be subject to prior approval by the Designer.

b) Reinforcing steel for bent and hooked connections shall conform to CSA CAN3-G40.21-M Grade 260W and shall be carefully bent to the radii detailed and installed as shown on the Standard SS Drawings.

Bending shall be done by methods that will not produce fracture or other injury. The metal heating shall not be to a higher temperature than that producing a "dark cherry red" colour. After heating, the metal shall be cooled as slowly as possible. Following the bending, the surface of the metal shall be carefully inspected for evidence of fracture, and any fractured pieces shall be replaced.

Contrary to Note 2 on SS Drawing SP941-04.01.01, prior to delivery, exposed surfaces of connections shall be prepared for and given a heavy application of zinc rich coating to CGSB Standard 1-GP-181M.

c) Pick-up points for handling units shall be formed with accurately placed rigid P.V.C. pipe recessed 15 mm from both finished surfaces as detailed.

941.04 Optional Features - Where barrier is ordered with drainage slots or grouting holes or both, they shall be accurately cast-in as detailed. Facilities for the installation of anti-glare screens will be detailed as and when required.

941.05 Placing and Finishing of Concrete

a) Concrete shall be placed in the forms and carefully consolidated in strict accordance with CSA CAN3-A23.4-M, Clause 19.

b) Curing and protection shall be carried out strictly according to CSA CAN3-A23.4-M Clause 21.

i) Curing shall be considered complete when test cylinders reach the specified 28 day compressive strength provided such strength is reached not later than 28 days after the barriers are cast.

ii) Steam curing is permissible for either the entire curing period or portion thereof and shall be carried out in accordance with CSA CAN3-A23.4-M

Clause 21.4.

iii) At no time during or at completion of the curing period shall the temperature differential between the concrete surface and the ambient temperature be greater than 20°C.

iv) If steam is used for a portion of the curing period, additional normal curing shall be carried out after the steam curing, according to CSA CAN3-A23.4-M Clause 21.3 until such time as strength tests of concrete test cylinders that have been both steam and normally cured with the barriers reach the specified concrete strength.

c) Exposed surfaces shall be uniform in texture and colour as produced from well maintained steel form surfaces and proper vibration methods without excessive surface fines or laitance.

d) Defects of the exposed surface will normally be cause for the rejection of any unit except where such are within the permissible limits or are subject to making good as follows:

i) Unobtrusive defects of any kind where their total area is not in excess of 2% of the exposed surface area of the unit.

ii) Air holes not greater than 3 mm in diameter and not more than 20 in any isolated 300 mm X 300 mm area.

iii) Sharp arises at the edges of the exposed surfaces where necessary shall be softened by careful rubbing or grinding.

iv) Patching of isolated small holes, cavities and similar self-confining defects may be permitted when authorized in writing by the Quality Manager.

e) Patching, only when authorized, shall have the defective area well saturated with water and the defect prepared with cement paste and filled with mortar. The mortar, properly proportioned of the same sand and cement as the original concrete and reasonably colour matched to the cured dry unit with the addition of white cement where necessary, shall be pre-shrunk for about one hour before retempering and use. The patching mortar shall be well tooled in, finished flush and smooth and the area covered to cure adequately.

f) End connection surfaces shall be cleared out.

g) All concrete surfaces prior to shipment shall be accurate to detail and, in particular at the end connections, true to dimension tolerances.

941.06 Tolerances - Tolerances allowable in concrete dimensions of the barriers shall be ± 3 mm except as otherwise indicated on the detail drawings.

941.07 Procedure of Manufacture - The Supplier shall notify the Contractor in advance concerning the date when the order is to be manufactured, so that detailed inspection can be carried out. All processes shall be open for inspection and approval by the Quality Manager and the Ministry Representative

The manufacturer's name or trade mark, year of manufacture and form number shall be embedded on the end of each unit in a manner, size and depth that they are permanently legible.

Authorized patching or making good may be inspected before shipment or upon delivery and the rejected unit(s) shall be replaced at no cost.

941.08 Handling - In handling the finished product, the concrete and connecting devices shall not be damaged or distorted.

SECTION 952

CONTRACTOR SUPPLY ASPHALT AND PAVING MATERIALS FOR HIGHWAY USE

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

952.01 Scope

a) This Section describes the requirements for the different types and grades of asphalt and paving materials to be supplied by the Contractor for highway use as set out in the Contract.

b) The materials so described are generally classified as follows:

- Asphalt Cements
- Cutback Asphalts
- Asphalt Emulsions
- Tall Oil Pitch
- Emulsified Products

c) The term "Supplier" referred to in this section shall mean the party or parties supplying the Contractor with materials covered under this specification.

d) The Ministry's *Recognized Products Book*, referenced herein, may be found on-line at:

http://www.th.gov.bc.ca/publications/eng_publications/geotech/rpb.htm

952.02 Quality Management Requirements

952.02.01 Quality Control Plan

a) The supplier shall develop and maintain a proven quality control plan in accordance with the provisions of this Specification and the Contract. The plan shall ensure that adequate inspection coverage is maintained throughout the entire process of manufacture and shipping. All supplies processed or manufactured within the supplier's plant or procured from any other source shall receive sufficient inspection to ensure conformance specifications, and evidence of such inspections shall be provided to the Quality Manager and Ministry Representative upon request.

b) Supplies not conforming to contractual requirements shall not be offered for highway use.

c) It shall be a responsibility of the Contractor to ensure that all materials supplied satisfy the related Environmental and Health and Safety Regulations.

d) Product discharged directly into tank cars or tank trucks from "in line blending" facilities will be sampled as required to ensure uniformity and satisfy quality requirements.

952.02.02 Quality Control Procedure - The supplier shall furnish the Contractor with an outline of the quality control procedures, for inclusion in the Quality Management Plan, detailing the method of implementing the requirements of this Specification. This outline shall include the following operations:

- sampling and testing
- storage and handling
- shipping
- recording and reporting.

952.02.03 Technical Requirements - The Contractor shall ensure that the supplier provides the following:

a) Batching and Batch Analysis: Each batch of product covered by this Specification shall be given a batch number, and sampled and tested in accordance with the applicable Specification.

The batch analysis must show the Specification requirements for the product tested, test method employed, and the pertinent test results, as listed on Appendix A.

b) A Viscosity Chart (Absolute) with the initial load and for each subsequent batch. If inline blending is utilized a new viscosity chart shall be issued should any change in viscosity occur.

952.03 Sampling and Testing

a) The supplier shall retain adequate sampling equipment, employ satisfactory sampling procedures, and maintain sampling program and records. The Contractor shall obtain samples for quality assurance.

b) Test methods for asphalt and paving materials shall conform to the standard ASTM tests listed in Table 952-A.

c) On request by the Ministry Representative, the Contractor shall provide samples, from the supplier, of the product in quantities, not less than 4 ℓ, as may be required for independent testing.

952.04 Delivery of Asphalt and Paving Materials

a) The Contractor must ensure the supplier delivers

asphalt and paving materials in good condition and at correct temperature to the specified delivery point

b) Every bill of lading must show:

- Type of product
- Batch number
- Mass
- Relative density at 15°C
- P.G. Specification (for information only)

c) Trucks shall be equipped with a submerged sampling valve system installed generally as shown on Drawing SS SP952-01 for each compartment in which asphalt material is to be carried out.

d) Delivery temperature of the shipment shall meet the viscosity requirement shown in Table 952-B.

e) Asphalt Emulsions and emulsified products shall be kept at temperatures above 5°C.

Note: In no case shall asphalt materials or Tall Oil Pitch be received at a viscosity outside the specified viscosity range.

TABLE 952-A TEST METHODS

TITLE OF TEST	TEST DESIGNATION	
		ASTM
Penetration		D 5
Absolute Viscosity of Asphalts		D 2171
Kinematic Viscosity of Asphalts		D 2170
Effect of Heat and Air on Asphalt		D 1754
Distillation of Cutback Asphalt		D 402
Residue of 100 Penetration		D 243
Relative Density of Asphalt Cement and Cutback Asphalt		D 70
Ductility		D 113
Float Test		D 139
Flash Point by Tag Open Cup		D 1310
Flash Point by Cleveland Open Cup		D 92
Solubility in Trichloroethylene		D 2042
Water in Asphalt		D 95
Softening Point in Ethylene Glycol		D 2398
Saybolt Furol Viscosity		D 88
Testing Emulsified Asphalts		D 244

TABLE 952-B VISCOSITY REQUIREMENTS FOR SHIPMENT DELIVERY TEMPERATURE

CLASS OF MATERIAL	KINEMATIC VISCOSITY, mm^2/s
Cutback Asphalt for Surface Spraying	100 - 200
Cutback Asphalt and Asphalt Cement for Plant Mixing	100 - 500
Tall Oil Pitch	20 - 40

952.05 Asphalt Cement - Asphalt cements shall:

a) Be products prepared by the refining of petroleum oils.

b) Be homogeneous, free from water and shall not foam when heated to 175°C.

c) Conform to the requirements specified in Table 952-C and Table 952-D.

Upon request, the Contractor shall forward a 4 ℓ sample of the product to the Ministry Representative.

NOTE: Values for penetration and viscosity are point values only. Intermediate values for Group A, B and C grades shall be obtained by interpolation between the tabulated values on a straight line basis. Minimum viscosity at 60°C for other penetrations within each group can be read from the corresponding straight line on Drawing SS SP952-02.

952.06 Cutback Asphalts

a) Cutback asphalts shall consist essentially of petroleum derivatives and shall be substantially free from water and other impurities.

b) Cutback asphalts shall be of the type and grade described in Tables 952-E to 952-H.

952.07 Asphalt Emulsions - Asphalt materials in the form of aqueous emulsions shall be of:

- Anionic Type
- Cationic Type
- High Float Type

952.08 Anionic Type Emulsion for Road Purposes - Anionic type of emulsion requirements are as follows:

- The asphalt emulsion shall be homogeneous.
- It shall show no separation of asphalt after thorough mixing within 30 days (crack filler 20 days) after delivery, provided separation has not been caused by freezing.
- The emulsion shall conform to the requirements listed in Table 952-I.

952.09 Cationic Type Emulsions for Road Purposes – Cationic type emulsions shall conform to the requirements listed in Table 952-J and shall be uniform throughout.

952.10 High Float Emulsified Asphalt - Storage stability - High float emulsified asphalt shall show no separation of asphalt within 30 days after delivery and shall be homogeneous after thorough mixing.
The emulsion shall conform to the requirements listed in Table 952-K.

952.11 Tall Oil Pitch (TOP) - Tall oil pitch, co-product of the pulp and paper industry is a residual fraction from the crude tall oil vacuum distillation process. It shall be a dark brown resinous and viscous material insoluble in water but miscible with asphalt. Where required, TOP will be added to asphalt cement in a ratio defined by the mix design, usually between 5 - 10%. Only TOP products listed on the Ministry's *Recognized Products Book* shall be permitted for use on Ministry projects.

The TOP shall conform to the requirements listed in Tables

TABLE 952-C GRADES OF ASPHALT CEMENT

PENETRATION GRADE ASPHALT CEMENT	80 - 100		150 - 200		200 - 300		TEST METHOD ASTM
Requirements	Min	Max	Min	Max	Min	Ma	
Penetration at 25°C 100 g and 5 s, 0.1 mm	80	100	150	200	200	300	D 5
Group A Group B Group C	Minimum viscosity values defining boundaries for groups A, B, & C are listed in Table 952-D and illustrated in Drawing SP952-04						
Flash Point, °C Cleveland Open Cup	230	---	220	---	175	---	D 92
Thin Film Oven Test % Loss in Mass	---	0.85	---	1.3	---	1.5	D 1754
Penetration of Residue at 25°C 100 g, 5 s, 0.1 mm, % of Original Penetration	55	---	50	---	45	---	After T.F.O.T.
Solubility in Trichloroethylene % by Mass	99.5	---	99.5	---	99.5	---	D 2042
Ductility, 25°C 5 cm/min, cm	100	---	100	---	---	---	D 113
Ductility, 15°C 5 cm/min, cm	---	---	---	---	100	---	

TABLE 952-D VISCOSITY AND PENETRATION VALUES DEFINING GROUP BOUNDARIES

Penetration 25°C	80	100	150	200	300	TEST METHOD - ASTM
						D 5
	Minimum Viscosity at 60°C					
Group A	150	115	70	50	31	D 2171
Group B	110	85	---	---	---	
Group C	75	55	---	---	---	

952-L and 952-M.

952.12 Emulsified Products - Emulsified products consist of emulsified Tall Oil Pitch and proprietary asphalt based emulsions. Only products on the Ministry's *Recognized Products Book* shall be permitted for use on the Ministry projects. Each batch of the product delivered to the job site shall be accompanied by the written statement from the manufacturer or supplier confirming that the product formulation complies with the original, previously approved formulation. Emulsified Tall Oil Pitch and asphalt based emulsified products shall conform to the requirements listed in Tables 952-N and 952-O.

TABLE 952-E SLOW CURING CUTBACK ASPHALT

REQUIREMENTS	GRADE					
	SC - 70		SC - 250		SC - 3000	
	Min	Max	Min	Max	Min	Max
Flash Point ^{o}C	65	---	80	---	105	---
Kinematic Viscosity at 60°C, mm^{2}/s	70	140	250	500	3 000	6 000
Residue from Distillation to 360°C, volume % by difference	70	90	80	94	95	---
Distillation Residue, Kinematic Viscosity at 60°C, mm^{2}/s	400	7 000	800	10 000	4 000	35 000
ASPHALT RESIDUE:						
Residue of 100 Penetration, % by Mass	50	---	60	---	80	---
Ductility of 100 Penetration residue at 25°C, cm	100	---	100	---	100	---
Solubility in Trichloroethylene	99.0	---	99.0	---	99.0	---
Water, % by Mass or Volume	---	0.5	---	0.5	---	0.5

TABLE 952-F MEDIUM CURING CUTBACK ASPHALT

REQUIREMENTS	GRADE			
	MC - 30		MC - 250	
	Min.	Max.	Min.	Max.
Flash Point ^{o}C	38	---	65	---
Kinematic Viscosity at 60°C, mm^{2}/s	30	60	250	500
Distillation Test, % of Total Distillate to 360°C: to 225°C	---	25	---	10
Distillation Test, % of Total Distillate to 360°C: to 260°C	40	70	15	55
Distillation Test, % of Total Distillate to 360°C: to 315°C	75	93	60	87
Residue from Distillation to 360°C, volume % by difference	50	---	67	---
PROPERTIES OF RESIDUE FROM DISTILLATION:				
Penetration at 25°C, 100 g, 5 s, 0.1 mm	120	250	120	250
Ductility at 25°C, cm	100	---	100	---
Solubility in Trichloroethylene	99.0	---	99.0	---
Water, %	---	0.2	---	0.2

TABLE 952-G RAPID CURING CUTBACK ASPHALT

REQUIREMENTS	GRADE					
	RC - 30		RC - 70		RC - 250	
	Min	Max	Min	Max	Min	Max
Flash Point ^{o}C	---	---	---	---	27	---
Kinematic Viscosity at 60°C, mm^{2}/s	30	60	70	140	250	500
Distillation Test, % of Total Distillate to 360°C: to 190°C	15	---	10	---	---	---
Distillation Test, % of Total Distillate to 360°C: to 225°C	55	---	50	---	35	---
Distillation Test, % of Total Distillate to 360°C: to 260°C	75	---	70	---	60	---
Distillation Test, % of Total Distillate to 360°C: to 315°C	90	---	85	---	80	---
Residue from Distillation to 360°C, volume % by difference	50	---	55	---	65	---
TEST ON RESIDUE FROM DISTILLATION:						
Penetration at 25°C, 100 g, 5 s, 0.1 mm	80	120	80	120	80	120
Ductility at 25°C, cm	100	---	100	---	100	---
Solubility in Trichloroethylene, % by mass	99.0	---	99.0	---	99.0	---
Water, %	---	0.2	---	0.2	---	0.2

Note: The material shall not foam when heated to the spraying and mixing temperature range recommended by the Canadian General Standards Board

TABLE 952-H CUTBACK ASPHALT PRIMER RM20

REQUIREMENTS	Min.	Max.
Kinematic Viscosity at 60°C, mm^{2}/s	20	35
Distillation Test, % of Total Distillate to 360°C: to 190°C	---	60
Distillation Test, % of Total Distillate to 360°C: to 225°C	40	---
Distillation Test, % of Total Distillate to 360°C: to 260°C	70	---
Distillation Test, % of Total Distillate to 360°C: to 315°C	85	---
Residue from Distillation to 360°C, volume % by difference	50	---
TEST ON RESIDUE FROM DISTILLATION:		
Penetration at 25°C, 100 g, 5 s, 0.1 mm	80	200
Ductility at 25°C, cm	100	---
Solubility in Trichloroethylene	99.0	---
Water, %	---	0.2

TABLE 952-I REQUIREMENTS FOR ANIONIC TYPE EMULSIONS

REQUIREMENT	TYPE OF EMULSION									
	RS - 1		RS - 2		MS - 2		SS - 1		CRACK FILLER	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Viscosity, Saybolt Furol: at 25°C	20	100	---	---	100	---	20	100	50	200
Viscosity, Saybolt Furol: at 50°C	---	---	75	300	---	---	---	---	---	---
Residue by Distillation, %	57	---	62	---	62	---	57	---	65	75
Settlement, 1 Day, %	---	1.5	---	1.5	---	1.5	---	1.5	---	1.5
Demulsibility: 50 mL of 0.1 N $CaCl_2$%	---	---	---	---	---	30	---	---	---	---
Demulsibility: 35 mL of 0.02 N $CaCl_2$%	60	---	60	---	---	---	---	---	30	---
Sieve Test, % Retained on 1 mm	---	0.10	---	0.10	---	0.10	---	0.10	---	0.10
Cement Mixing Test, %	---	---	---	---	---	---	---	2.0	---	---
TESTS ON RESIDUE:										
Penetration at 25°C, 100 g, 5 s	100	200	100	200	100	200	100	200	60	100
Solubility in Trichloroethylene %	97.5	---	97.5	---	97.5	---	97.5	---	97.5	---
Ductility at 25°C, cm	60	---	60	---	60	---	60	---	60	---

TABLE 952-J REQUIREMENTS FOR CATIONIC TYPE EMULSIONS

REQUIREMENT	TYPE OF EMULSION			
	CRS - 1K		CRS - 2K	
	Min	Max	Min	Max
Saybolt Furol Viscosity at 50°C	30	125	174	400
% Residue by Distillation	62	---	68	---
Settlement 1 Day, %	---	1.5	---	1.5
Sieve Test, % Retained on 1 mm Mesh	---	0.1	---	0.1
Oil Portion of Distillate, % of Total Volume	0	3	0	3
Particle Charge	POSITIVE		POSITIVE	
TESTS ON RESIDUE				
Penetration at 25°C, 100 g, 5 s	100	250	100	150
Solubility in Trichloroethylene %	97.5	---	97.5	---
Ductility at 25°C, cm	60	---	65	---

TABLE 952-K REQUIREMENTS FOR HIGH FLOAT EMULSIFIED ASPHALTS

REQUIREMENT	GRADE:											
	HF-100S		HF-150S		HF-250S		HF-350S		HF-500M		HF-1000M	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Residue by Distillation, % By Mass	62	---	62	---	62	---	65	---	65	---	65	---
Oil Distillate % By Volume	1	4	0.5	4	1	6	1.5	6	1	6	1	7
Saybolt Viscosity, Furol Seconds at 50°C	35	150	35	150	35	150	75	400	50	---	50	---
Sieve Test, % Retained on 1 mm Sieve	---	0.1	---	0.1	---	0.1	---	0.1	---	0.1	---	0.1
Coating Test %	90	---	90	---	90	---	---	---	---	---	---	---
Settlement 1 Day, % By Mass	---	1.5	---	1.5	---	1.5	---	1.5	---	1.5	---	1.5
Demulsibility: 50 mL 5.55 g/L $CaCl_2$, % By Mass	75	---	75	---	---	---	---	---	---	---	---	---
Workability @ 10°C	---	---	---	---	---	---	---	---	---	---	Pass	---
TEST ON RESIDUE												
Penetration at 25°C, 100 g, 5 s	*		**		**		**		---	---	---	---
Viscosity at 60°C, Pa·s	*		**		**		**		8	20	2	8
Float Test at 60°C, s	1200	---	1200	---	1200	---	1200	---	1200	---	1200	---
Solubility in Trichloroethylene, %	97.5	---	97.5	---	97.5	---	97.5	---	97.5	---	97.5	---

* See Drawing SS SP952-02 ** See Drawing SS SP952-03

TABLE 952-L TALL OIL PITCH VISCOSITY AND PENETRATION DATA

REQUIREMENTS	Min.	Max.
Results on original sample		
Absolute Viscosity, 60°C, PaCs	1	2
Kinematic Viscosity, 135°C, mm^2/s	25	35
Penetration, 4°C/100g, 5 sec, 0.1 mm	150	250
Results after Thin Film Oven		
Loss in weight, %	---	0.65
Absolute Viscosity, 60°C, PaCs	---	3
Kinematic Viscosity, 135°C, mm^2/s	---	60
Penetration, 4°C, 100 g, 5 sec, 0.1 mm	75	---

TABLE 952-M TALL OIL PITCH DATA

REQUIREMENTS		Min.	Max.
Softening Point, °C	(ASTM D 36)	---	35
Flash Point, Cleveland Open Cup, °C	(ASTM D 92)	250	---
Fire Point, Cleveland Open Cup, °C	(ASTM D 92)	275	---
Boiling Point, °C		320	---
Specific Gravity		0.94	0.98
Vapour Pressure, mm Hg		---	1
PH		3.75	4.25
Wood Extractive, %		98	---
Ash, %	(ASTM D 803)	---	0.8
Moisture, %	(ASTM D 803)	---	0.1
Fatty Acids, %		7	9
Resin Acids, %		5	7
Unsaponifiables, %		39	44
Neutrals, %		42	46
Acid Number		20	30

TABLE 952-N EMULSIFIED TALL OIL PITCH

REQUIREMENTS	Min.	Max.
Viscosity, SF, 25°C, s	10	30
Residue by Distillation, % by weight	40	---
Oil Distillate, % by volume	---	0.1
Settlement 24hrs, %	---	1.5
pH	6	8
Particle Charge	Negative	
Miscibility with Water	Pass	
Specific Gravity, 20°C	Approx. 1.0	
Boiling Point, °C	100	
Freezing Point, °C	0	
Vapour Pressure, mm Hg	20	
Odour and Appearance - distinctive resinous Odour, light yellow colour		
TEST ON RESIDUE - Shall conform to the requirements for Tall Oil Pitch listed in table 951-L		

TABLE 952-O EMULSIFIED PRODUCTS

REQUIREMENTS	Min.	Max.
Viscosity, SF , 25°C, s	10	60
Residue by Distillation, % by weight	40	---
Oil Distillate, % by Volume	---	5
Settlement 24hrs, %	Pass	
Miscibility with Water	Pass	
TEST ON RESIDUE		
Penetration, 25°C, 100 g, 5 sec, 0.1 mm	40	200

APPENDIX A

The following data is required from the supplier:

ASPHALT CEMENTS

Grade and Type (when required)

A. Original Asphalt

Density @ 15°C, kg/L

- Viscosity @ 135°C
- Viscosity @ 60°C
- Penetration @ 25°C, 100 g/5 s
- Solubility in Trichloroethylene
- Ductility @ 25°C
- Flash Point C.O.C.
- % Retained Penetration after T.F.O.T. @ 25°C

B. Residue After T.F.O.T.
(when Type A required)

- Penetration @ 25°C
- Penetration @ 10°C
- Penetration @ 4°C

NOTE: Viscosity - Temperature chart also required for each batch.

CUTBACK ASPHALTS

A. Cutback Asphalts as Supplied

- Flash Point, T.O.C. (C.O.C. above 79°C)
- Density @ 15°C, g/cm^3
- Viscosity @ 60°C, mm^2/s
- Distillate, % by volume of Total
  - Distillate to 360°C
  - Distillate to 190°C
  - Distillate to 225°C
  - Distillate to 260°C
  - Distillate to 315°C
- Residue from Distillation to 360°C
- Volume % by Difference

B. Properties of Residue

- Penetration @ 25°C
- Ductility @ 25°C
- Solubility in Trichloroethylene, %

C. Residue of 100 Penetration by Mass

- Ductility of 100 Penetration at 25°C, cm

EMULSIONS

A. Original Emulsion

- Saybolt Furol Viscosity @ 25°C
- Saybolt Furol Viscosity @ 50°C
- % Residue by Distillation
- Settlement in 5 Days, %
- Oil Portion of Disillate
- % to Total Volume

B. Tests on Residue

- Float Test @ 60°C, s
- Penetration @ 25°C, 100 g/5 s
- Ductility @ 25°C

SECTION 991

CALCIUM CHLORIDE AND SODIUM CHLORIDE

991.01 Calcium Chloride - Calcium chloride, for highway purposes, shall conform to CGSB Specification 15-GP-1M Calcium Chloride. It shall be supplied as Type I Regular (77%) bulk (flake) or sacked, or as liquid at 35% solution concentration, as specified.

991.02 Sodium Chloride - Sodium chloride, for highway purposes, shall conform to CGSB Specification 15-GP-9M Sodium Chloride, Pavement De-icer but it shall be supplied as shown in Table 991-A.

991.03 Chemical Contamination of Road Salt - When applied as specified, the de-icing chemicals shall not contain substances that exceed the current environmental guidelines, Provincial or Federal, and shall not cause harm to flora and fauna.

Note: CGSB Specifications are obtainable from:
Canadian Government Specification Board,
Ottawa, ON K1A 0S5

TABLE 991-A GRADATION FOR SODIUM CHLORIDE

SCREEN SIZE	PERCENT PASSING
12.0 mm (1/2")	100
9.00 mm (3/8")	90 - 100
4.75 mm (#4)	35 - 85
2.40 mm (#8)	15 - 55
1.20 mm (#16)	5 - 30
0.600 mm (#30)	0 - 10